

VIVEKANAND EDUCATION SOCIETY'S INSTITUTE OF TECHNOLOGY

**(An Autonomous Institute Affiliated to University of Mumbai
Department of Computer Engineering)**

Department of Computer Engineering



Project Report on

DeBug : A Decentralised Bug-Bounty Platform

In partial fulfilment of the Fourth Year (Semester VII), Bachelor of Engineering
(B.E.) Degree in Computer Engineering at the University of Mumbai,
Academic Year 2025-26

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Submitted by

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(2025-26)

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CERTIFICATE

This is to certify that **Rohit Shahi [D17C/58]**, **Gopal Vanjarani [D17C/66]**, **Gaurav Mahadeshwar [D17C/43]** and **Umesh Tolani [D17C/64]** of Fourth Year Computer Engineering studying under the University of Mumbai has satisfactorily presented the project on “**DeBug : A Decentralised Bug-Bounty Platform**” as a part of the coursework of PROJECT-I for Semester-VII under the guidance of **Ms. Geocey Shejy** in the year 2024-2025.

Date

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Dr. J. M. Nair

Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea / data / fact / source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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We wish to express our profound thanks to all those who helped us in gathering information about the project. Our families too have provided moral support and encouragement several times.

Computer Engineering Department

COURSE OUTCOMES FOR B.E PROJECT

Learners will be to:-

Course Outcome	Description of the Course Outcome
CO1	Do literature survey/industrial visit and identify the problem of the selected project topic.
CO2	Apply basic engineering fundamentals in the domain of practical applications for problem identification, formulation and solution
CO3	Attempt & Design a problem solution in the right approach to complex problems
CO4	Cultivate the habit of working in a team
CO5	Correlate the theoretical and experimental/simulation results and draw the proper inferences
CO6	Demonstrate the knowledge, skills and attitudes of a professional engineer, & prepare a report as per the standard guidelines.

Abstract

DeBug – A Decentralized Bug-Bounty Platform is an innovative blockchain-powered system designed to redefine the way organizations and security researchers collaborate for vulnerability reporting and resolution. The platform integrates decentralized architecture, smart contracts, and cryptographic technologies to ensure transparency, fairness, and automation throughout the bug-bounty lifecycle. DeBug leverages Solana-based smart contracts for secure escrow management, guaranteeing verifiable and tamper-proof payouts upon successful vulnerability validation. The hybrid design incorporates on-chain and off-chain components, where the blockchain layer handles financial and trust mechanisms, while the off-chain infrastructure supports encrypted vulnerability submissions, authentication, and real-time notifications.

A key feature of DeBug is its privacy-preserving architecture that employs encrypted communication channels and Zero-Knowledge (ZK) identity verification, allowing researchers to maintain anonymity while ensuring the authenticity of submissions. The integrated reputation system records researcher performance and credibility across programs, enabling reputation portability and reducing reliance on centralized intermediaries. Additionally, DeBug's governance framework supports dispute resolution and transparent audits through decentralized consensus, eliminating bias and ensuring fair handling of reports.

By combining the immutability of blockchain with the scalability of traditional web systems, DeBug creates a trustless, transparent, and efficient vulnerability management ecosystem. The platform empowers organizations to strengthen their security posture while rewarding ethical hackers in a fair and timely manner. Future enhancements include multi-chain compatibility, DAO-driven governance, and integration of advanced analytics to improve triage efficiency and researcher engagement. In essence, DeBug bridges the gap between centralized bug-bounty programs and a decentralized, privacy-first, globally accessible cybersecurity marketplace.

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Chapter 1: Introduction

1.1 Introduction

Cybersecurity has become a critical component of modern digital infrastructure, ensuring the protection of systems, data, and user trust. Bug-bounty programs play a key role in this ecosystem by enabling ethical hackers to identify and report vulnerabilities before they are exploited. However, most existing platforms are centralized, leading to issues such as delayed disclosures, opaque payouts, custodial risks, and privacy concerns. These challenges reduce researcher motivation and limit the overall efficiency of vulnerability management.

With the emergence of blockchain and smart contract technologies, decentralization offers a promising solution to address these limitations. DeBug introduces a hybrid decentralized bug-bounty platform that combines blockchain transparency with the flexibility of off-chain services. The system uses Solana-based smart contracts for automated and verifiable payouts, while off-chain modules handle authentication, encrypted submissions, and notifications.

DeBug also integrates Zero-Knowledge (ZK) identity verification to ensure researcher privacy and a decentralized reputation system to enable credibility and recognition across platforms. By merging transparency, privacy, and scalability, DeBug creates a trustless, efficient, and globally accessible ecosystem that redefines how vulnerabilities are reported, validated, and rewarded.

1.2 Motivation

In today's digital era, cybersecurity has become a fundamental requirement for every organization, yet the increasing number of cyberattacks and data breaches highlights the limitations of existing vulnerability management systems. Traditional bug-bounty platforms, though effective in theory, often face challenges such as delayed payouts, opaque dispute resolution, and lack of researcher privacy. Moreover, centralized control over submissions and reward processes introduces custodial risks, discouraging many ethical hackers from participating.

The motivation behind developing DeBug stems from the need to create a transparent, trustless, and globally accessible ecosystem for vulnerability disclosure. By leveraging blockchain technology, DeBug ensures verifiable payouts, decentralized governance, and tamper-proof audit trails, thereby eliminating the dependence on intermediaries. The integration of smart contracts on Solana allows automated bounty distribution, ensuring fairness and efficiency in researcher compensation.

Another driving factor is the growing demand for privacy-preserving participation. Through Zero-Knowledge (ZK) identity verification and encrypted submissions, DeBug enables anonymous yet credible reporting, protecting the identity and reputation of security researchers. The platform's decentralized reputation system further motivates consistent, ethical participation by allowing recognition and credibility to be carried across multiple programs.

Ultimately, DeBug aims to democratize cybersecurity collaboration—empowering organizations to strengthen their digital defenses while enabling researchers worldwide to contribute securely, transparently, and fairly in a decentralized environment.

1.3 Problem Definition

Despite advancements in cybersecurity platforms, existing bug-bounty systems remain limited in terms of transparency, privacy, and efficiency. Current centralized solutions often fail to provide fairness in payouts, consistent triage, or anonymity for researchers. These shortcomings create trust gaps between security researchers and organizations.

Existing platforms suffer from the following limitations:

- **Centralization and Trust Issues:** Platforms act as intermediaries controlling submissions and payouts, leading to delayed disclosures, disputes, and lack of transparency.
- **Incentive Misalignment:** Low or capped payouts often discourage ethical hackers, making vulnerability reporting less rewarding compared to exploitation.
- **Payment Friction:** Reliance on fiat-based payments causes delays, high fees, and limited support for global participants.

- Privacy Limitations: Mandatory KYC and real-identity verification compromise researcher anonymity, discouraging participation.
- Opaque Dispute Handling: Dispute resolutions are managed privately by centralized administrators, leading to potential bias and reduced credibility.
- Lack of Reputation Portability: Researcher performance and recognition are confined to individual platforms, restricting global acknowledgment.

Hence, the core problem addressed by this project is:

To design and develop a decentralized, blockchain-based bug-bounty platform that ensures transparent payouts through smart contracts, protects researcher privacy via encrypted submissions and zero-knowledge identity verification, and provides a scalable, trustless environment for fair and efficient vulnerability management.

1.4 Relevance of the Project

DeBug holds strong relevance in the evolving field of cybersecurity, where transparency, trust, and automation are becoming essential to effective vulnerability management. With the increasing number of cyber threats and the global shift toward open collaboration between researchers and organizations, decentralized platforms present a transformative solution. By leveraging blockchain and smart contracts, DeBug provides a transparent, scalable, and privacy-preserving alternative to conventional bug-bounty systems, aligning with global trends in decentralized governance, Web3 security, and ethical hacking.

From an academic and technological standpoint, the project contributes to the domains of Blockchain Technology, Smart Contracts, Cryptography, and Web Security, offering practical insights into their integration for real-world problem-solving. It supports the broader goals of enhancing trust in digital ecosystems and promoting fair, transparent collaboration between stakeholders in cybersecurity. Furthermore, the project aligns with the United Nations Sustainable Development Goal (SDG) 9 – Industry, Innovation, and Infrastructure, by promoting secure, innovative, and decentralized technological infrastructure.

In summary, DeBug's relevance is defined by its potential to:

- Strengthen trust and transparency in vulnerability disclosure through blockchain verification.
- Ensure fair and automated reward distribution using smart contracts.
- Protect researcher privacy through encrypted submissions and zero-knowledge identity verification.
- Promote global participation in cybersecurity by removing central intermediaries and accessibility barriers.

1.5 Methodology Used

The development methodology for DeBug is organized into interconnected stages designed to ensure transparency, security, and scalability across the entire bug-bounty lifecycle. The system adopts a hybrid approach, combining blockchain smart contracts with off-chain services to maintain efficiency while leveraging decentralization for trust and automation.

Step 1: Bounty Creation

The process begins when the organization drafts a bounty with defined tiers, severity levels, and deadlines. An escrow smart contract is deployed on the Solana blockchain to securely hold the reward amount. This ensures verifiable fund allocation before any vulnerability submission.

Step 2: Submission

Researchers prepare and encrypt their vulnerability reports on the client side to preserve confidentiality. Reports are stored on IPFS (InterPlanetary File System), ensuring immutability and decentralization, while essential metadata and pointers are maintained in a PostgreSQL database via the backend.

Step 3: Validation and Triage

The backend, developed using Express.js and Prisma ORM, validates the submission format and authenticity. The organization's security team reviews vulnerabilities through a secured auditing interface, ensuring only legitimate and impactful issues are accepted for payout consideration.

Step 4: Decision and Payout

Once a vulnerability is approved, the Solana smart contract automatically releases the reward to the researcher's connected crypto wallet. Disputes, if any, are managed through an off-chain arbitration service linked with on-chain contract hooks for transparency and fairness.

Step 5: Settlement and Reputation

After payout, the smart contract emits a verifiable transaction event. The backend updates the researcher's reputation score, maintaining a decentralized record of credibility across all programs. Notifications, analytics, and performance logs are updated in real-time, enabling users to track outcomes seamlessly.

This multi-step methodology integrates blockchain transparency, cryptographic privacy, and efficient off-chain processing, creating a robust, verifiable, and scalable ecosystem for ethical vulnerability management.

Chapter 2: Literature Survey

2.1 Research Papers Referred

Publication Year	Title of Paper	Abstract	Research Gap	Conclusion
2021	Bountychain: A Decentralized Bug Bounty Platform (Hoffman et al.)	Proposes a blockchain-based bug bounty platform using Ethereum smart contracts and IPFS to automate payouts and store immutable records.	Limited by Ethereum gas fees and on-chain storage constraints for large reports.	Demonstrates feasibility of decentralized bug bounty programs; recommends optimizing cost and scalability.
2020	VeriOSS: Using the Blockchain to Foster Bug Bounty Programs	Introduces VeriOSS, a blockchain-based platform ensuring fairness and trust in bug disclosure via economic incentives and formal guarantees.	Limited to local software copies; lacks exploit support and robust identity verification; assumes single participants.	Promotes trustless, fair rewards; needs refinement for broader applicability and security.
2023	Predicting the Effectiveness of Blockchain Bug Bounty Programs	Analyzes ~200 blockchain bug bounty programs to identify success factors and participation trends.	Missing/uneven data across platforms; limited variables.	Clear guidelines and documentation improve participation; future work to refine model and dataset.
2019	An Empirical Study of Bug Bounty Programs	Examines economics of traditional bug bounty programs across HackerOne and BugCrowd.	Limited transparency and productivity data; lacks longitudinal models.	Bug bounties are cost-effective complements to in-house security efforts.

Table no. 01 - Research papers referred.

2.2 Existing Systems

Feature	Bugcrowd	Immunefi	Hats Finance	Feature
Type	Centralized Bug Bounty Platform	Web3/DeFi Security Bug Bounty Platform	DAO-based Decentralized Bug Bounty Protocol	Type
Primary Focus	Traditional web, mobile, and cloud security	Smart contracts, DeFi, blockchain infrastructure	Blockchain security and community-driven bounty vaults	Primary Focus
Governance Model	Managed by Bugcrowd team	Managed by Immunefi team	Decentralized via DAO (governance token: HAT)	Governance Model
Report Handling	Private submission via platform UI	Private submission; project verifies validity	Encrypted submission, revealed post-validation by DAO	Report Handling

Table no. 02 - Existing Systems

2.3 Lacuna in Existing Systems

While existing systems demonstrate the growing influence of AI in interview preparation and recruitment, several gaps remain unaddressed:

1. Centralization & Trust

Existing bug-bounty platforms operate under centralized control, leading to disputes, delayed disclosures, and lack of transparency. This dependency on intermediaries reduces trust among researchers and organizations.

2. Incentive Misalignment

Many platforms offer low or capped rewards, discouraging ethical hackers. The imbalance between effort and payout makes vulnerability disclosure less appealing compared to exploitation.

3. Payment Friction

Traditional fiat-based payments introduce delays, high transaction fees, and limited crypto support, making global participation slow and inconvenient.

2.4 Comparison of Existing Systems and Proposed Work

Sr. No	Existing Systems	Proposed Work (DeBug)
1	Existing platforms such as Bugcrowd and HackerOne operate on centralized architectures, where the platform controls report handling, payouts, and dispute resolution. This creates custodial risks, delayed payments, and lack of transparency.	DeBug eliminates intermediaries by using Solana-based smart contracts for automated escrow and payouts, ensuring verifiable and tamper-proof financial transactions.
2	Platforms like Immunefi and Hats Finance focus mainly on Web3 projects but rely on project-level verification and semi-centralized management, which can introduce bias in validation.	DeBug employs a hybrid on-chain/off-chain architecture , where validation is handled transparently through backend auditing services while maintaining blockchain-based payout logic.
3	Existing systems often require real identity verification (KYC), limiting anonymity and exposing researchers to privacy risks.	DeBug integrates Zero-Knowledge (ZK) identity verification and encrypted submissions on IPFS, allowing pseudonymous participation while ensuring report authenticity.
4	Payment systems in traditional platforms depend on fiat or manual crypto transfers, leading to high fees and delayed rewards.	DeBug automates the payment process via smart contract-triggered crypto payouts , offering instant and fee-efficient settlements.
5	Current platforms lack a unified, cross-platform reputation system, confining researcher credibility to a single platform.	DeBug introduces a decentralized reputation mechanism that tracks verified contributions and allows reputation portability across multiple bounty programs.

Table No. 03 - Comparison of Existing Systems and Proposed Work

2.5 Focus Area

The primary focus area of DeBug is to design and implement a hybrid decentralized bug-bounty platform that ensures transparent, secure, and privacy-preserving collaboration between organizations and security researchers. The system emphasizes blockchain-driven automation, encrypted vulnerability submissions, and verifiable payouts through smart contracts to eliminate trust issues and central dependencies.

By integrating Solana-based smart contracts, IPFS for decentralized storage, and Zero-Knowledge (ZK) identity verification, DeBug provides a scalable and efficient mechanism for ethical vulnerability disclosure. The project also focuses on building a decentralized reputation system that enables credibility tracking and recognition portability across programs.

The central aim of DeBug is to create a trustless, globally accessible, and researcher-friendly cybersecurity ecosystem—one that ensures fairness, enhances transparency, protects privacy, and fosters responsible vulnerability reporting through the seamless convergence of blockchain technology and modern web infrastructure.

Chapter 3: Requirements

3.1 Proposed model

The DeBug platform follows a hybrid decentralized model that integrates blockchain-based transparency with off-chain scalability and secure data handling. It is designed for trustless collaboration, privacy preservation, and automation in the bug-bounty ecosystem.

The proposed architecture is divided into three main layers — Presentation Layer, Application Layer, and Blockchain & Data Layer.

The Presentation Layer, built with React.js and Tailwind CSS, serves as the user interface where researchers and organizations interact with the system. It includes pages for bounty creation, profile management, submission tracking, and analytics dashboards.

The Application Layer, developed using Node.js and Express.js, acts as the bridge between the frontend and the blockchain. It manages API requests, authentication, report validation, and database communication. This layer ensures smooth interaction between users and blockchain components through secure RESTful APIs and MetaMask wallet integration.

The Blockchain & Data Layer is the system's backbone. It uses Solana-based smart contracts for secure escrow management and automatic payout disbursement. Vulnerability reports are stored on IPFS (InterPlanetary File System) to ensure decentralized and tamper-proof data storage, while metadata, user details, and analytics are maintained in PostgreSQL for efficient retrieval.

This layered design ensures that the blockchain handles trust and transparency, while off-chain components manage scalability, efficiency, and user experience — creating a balanced, privacy-focused, and verifiable ecosystem.

3.2 Functional Requirements

The functional requirements of DeBug define the core operations and expected behavior of the system to ensure secure, automated, and transparent bug-bounty management.

Key Functional Requirements:

- User Registration and Wallet Authentication – The system must allow users (researchers and organizations) to register securely and log in using their MetaMask wallet.
- Bounty Creation and Configuration – Organizations should be able to create, update, and close bounties while defining scope, reward tiers, and deadlines.
- Encrypted Report Submission – Researchers must be able to submit vulnerability reports in an encrypted format to ensure privacy and data protection.
- Smart Contract-Based Escrow – Each bounty must be linked to a Solana smart contract that locks funds until a report is validated.
- Report Validation and Triage – The backend should validate submission format, severity, and authenticity before forwarding it for review.
- Automated Payout – Upon approval, the smart contract automatically releases the reward to the researcher's wallet.
- Dispute Resolution – In case of conflicts, the system should enable off-chain arbitration connected to on-chain contract hooks for fair judgment.
- Reputation System – Each researcher should have a verifiable reputation score that updates after each valid submission.
- Analytics and Dashboard – Admin and users must be able to view bounty statistics, payment history, and submission trends.
- Community Governance – DAO-based voting should be enabled for community participation in triage and validation decisions.
- Cloud Deployment and Web Access – The platform should be deployable on cloud infrastructure and accessible through any web browser.

3.3. Non-Functional Requirements

- **Performance:** Smart contract execution and database queries must be optimized for minimal latency and cost efficiency.
- **Usability:** The platform should provide an intuitive UI and smooth navigation for both technical and non-technical users.
- **Reliability:** The system must ensure consistent smart contract behavior and secure communication between blockchain and backend.
- **Scalability:** The architecture should support multiple bounties, large datasets, and a growing user
- **Security:** Implement encryption for all submissions, secure wallet authentication, and data integrity checks for every transaction.
- **Transparency:** Every transaction, payout, and decision should be verifiable through on-chain logs and immutable records.
- **Maintainability:** The modular architecture should allow easy updates to smart contracts, APIs, and frontend components.

3.4. Hardware & Software Requirements

The development and deployment of DeBug require a well-balanced combination of efficient hardware infrastructure and robust software tools to ensure decentralized functionality, transparency, and secure operation across blockchain and web layers.

Hardware Requirements

- **Development System:** A laptop/PC with a minimum Intel i5 / AMD Ryzen 5 processor (x64 architecture).
- **Memory (RAM):** Minimum 8 GB (16 GB recommended for smoother backend and blockchain interactions).
- **Storage:** Minimum 50 GB free storage (SSD preferred for faster build and local node operations).
- **Graphics Support:** Basic GPU support (optional) for running encryption and Dockerized environments efficiently.

- Internet Connectivity: Stable broadband connection required for blockchain RPC calls, IPFS interactions, and API communication.
- Deployment Server: Cloud-based environment (AWS ECS / Render / Vercel) with 4 vCPUs, 8 GB RAM, and Docker/Kubernetes support for container orchestration.

Software Requirements

- Operating System: Windows 10/11, macOS, or Ubuntu Linux (latest LTS version).
- Frontend: Next.js (React framework) for a responsive, modular user interface.
- Backend: Node.js with Express, TypeScript, and Redis for secure and optimized server-side operations.
- Blockchain Layer: Solana / Polygon SDK for deploying and interacting with smart contracts managing escrow and payout transactions.
- Storage Layer: NeonDB serverless PostgreSQL for structured metadata, integrated with IPFS for decentralized file and report storage.
- Identity & Security: RainbowKit, JWT authentication, and Zero-Knowledge (ZK) proof for secure, pseudonymous access and identity verification.
- Infrastructure Tools: Docker and Docker Compose for containerized microservices, with Prometheus and Grafana for system monitoring and performance metrics.
- CI/CD Pipeline: GitHub Actions, Vercel, and Render for version control, continuous integration, and deployment automation.
- Development Tools: Visual Studio Code IDE, Postman for API testing, and Prisma ORM for database modeling and schema management.

This configuration ensures that DeBug runs efficiently across all environments — from development and testing to decentralized deployment — providing a reliable, scalable, and secure infrastructure for blockchain-based bug-bounty management.

3.5 Constraints of Working System

Despite its modular and decentralized design, the current working version of DeBug faces certain constraints typical to blockchain-integrated platforms:

- **Blockchain Transaction Costs:** Smart contract deployment and bounty payout transactions incur variable gas fees depending on network congestion and the underlying blockchain (Solana/Polygon).
- **API and Network Dependency:** The backend relies on third-party blockchain RPC endpoints, IPFS nodes, and gateway APIs. Downtime or latency in these services can impact real-time performance.
- **Continuous Network Requirement:** Stable internet connectivity is essential for wallet authentication, on-chain transactions, and decentralized data access.
- **High Computational Load:** Cryptographic operations, smart contract validation, and encryption algorithms demand higher CPU usage during active sessions.
- **Operational Costs:** Running cloud-based servers, maintaining Dockerized services, and hosting monitoring tools such as Grafana and Prometheus introduce recurring costs.
- **Data Privacy and Compliance:** Although reports are encrypted and pseudonymous, they must adhere to international data security and compliance standards such as GDPR.
- **Limited Development Time:** Due to the academic scope of the project, advanced features such as DAO voting integration, automated severity analysis, and multi-chain interoperability are scheduled for future iterations.

Chapter 4: Proposed Design

4.1 Block Diagram (InterviewIQ)

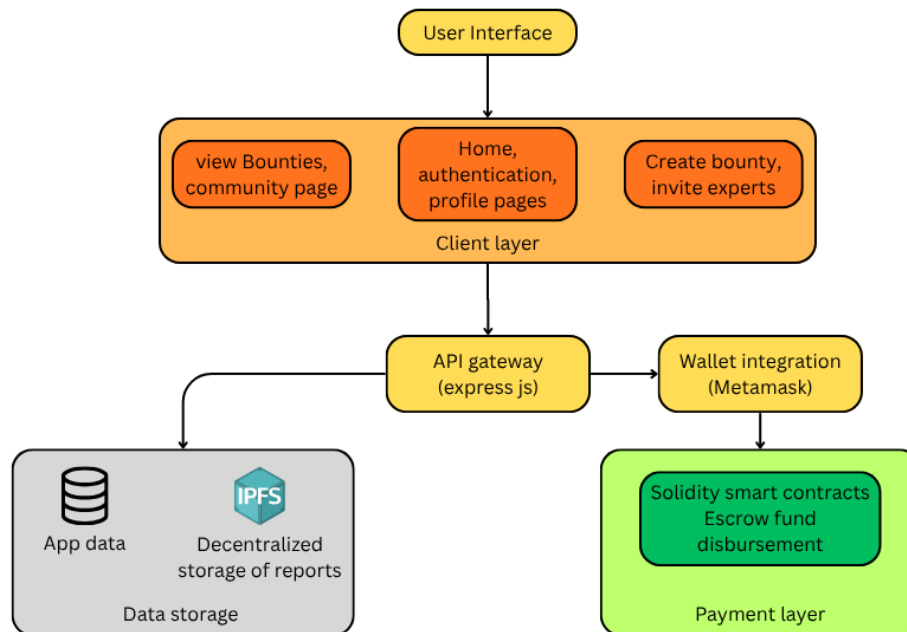


Fig No. 01 - Block Diagram

The above block diagram represents the overall system architecture of **DeBug**, illustrating how various on-chain and off-chain components interact to enable a secure, transparent, and efficient decentralized bug-bounty process.

User Interface Layer:

This is the primary interaction layer for users, including both organizations and researchers. It provides access to features such as viewing bounties, managing profiles, creating new bounty programs, and inviting security experts. The interface is designed to be intuitive and responsive, ensuring seamless user interaction across all devices.

Client Layer:

Built with modern web technologies, this layer handles all visual elements and user-side operations. It includes pages such as the community page, authentication portal, and bounty

management dashboard. The client layer ensures smooth navigation and communication with backend services via secure APIs and wallet integrations.

API Gateway (Express.js):

The API Gateway serves as the bridge between the frontend and backend. Developed using **Express.js**, it manages all API requests, performs user authentication, and coordinates communication between the blockchain network, decentralized storage, and the database. It also handles verification logic and report triage for vulnerability submissions.

Wallet Integration (MetaMask):

This component allows researchers and organizations to connect their **MetaMask wallets** for authentication and transactions. Through wallet integration, users can create bounties, stake funds, and receive payouts directly from the blockchain, ensuring transparency and eliminating third-party dependency.

Payment Layer (Smart Contracts):

At the core of the system lies the **Solidity smart contract**, deployed on the **Solana blockchain**. It governs the escrow mechanism, automatically holding and disbursing bounty funds upon successful validation of vulnerabilities. The smart contract emits verifiable on-chain events for each payout, ensuring immutable transaction records and trustless execution.

Data Storage:

DeBug employs a hybrid data storage approach combining **IPFS** and a traditional backend database. Sensitive bounty reports are stored in a decentralized manner using IPFS for immutability and security, while metadata, user logs, and analytics are maintained in the backend database. This ensures scalability, data persistence, and privacy for all stakeholders.

Together, these layers create a **hybrid decentralized architecture** that combines the transparency of blockchain with the efficiency of off-chain computation. The result is a secure, privacy-preserving, and scalable bug-bounty ecosystem that ensures fairness, automation, and trust across all transactions.

4.2 Modular diagram

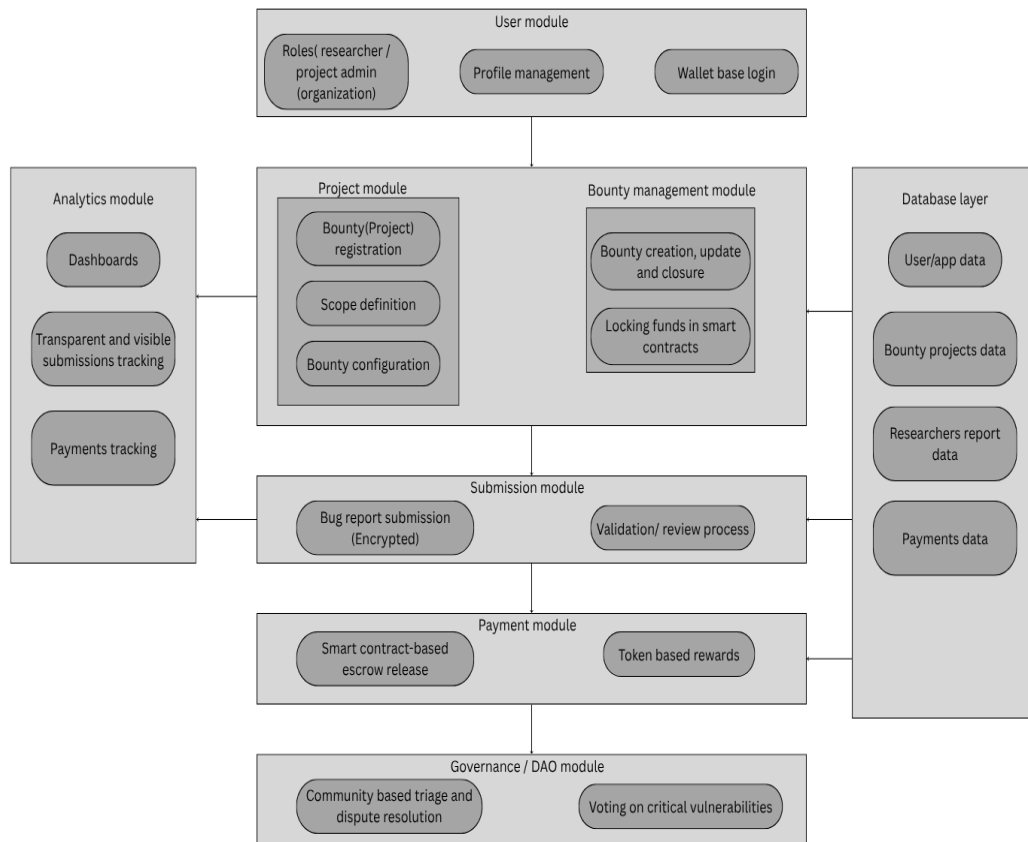


Fig No. 02 - Modular Diagram

The modular architecture of DeBug illustrates the complete workflow of the decentralized bug-bounty system, showing how each component contributes to secure and transparent vulnerability management. The User Module manages wallet-based authentication and profile handling for researchers and organizations. The Project and Bounty Management Modules handle bounty registration, scope definition, fund locking, and configuration through smart contracts on the blockchain. The Submission Module manages encrypted bug report submissions and validation, ensuring privacy and authenticity. The Payment Module automates token-based rewards and escrow releases, while the Governance (DAO) Module facilitates decentralized voting, triage, and dispute resolution. Supporting modules like Analytics provide dashboards and real-time tracking, and the Database Layer securely stores user, bounty, and payment data. Together, these modules ensure a transparent, scalable, and trustless bug-bounty ecosystem.

4.3 Detailed Design

DFD Diagram - Level 0

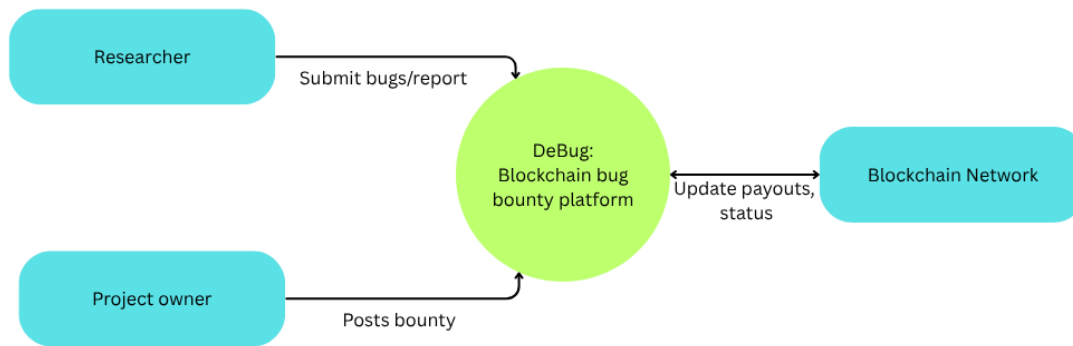


Fig No. 03 - DFD Level-0 Diagram

DFD Diagram - Level 1

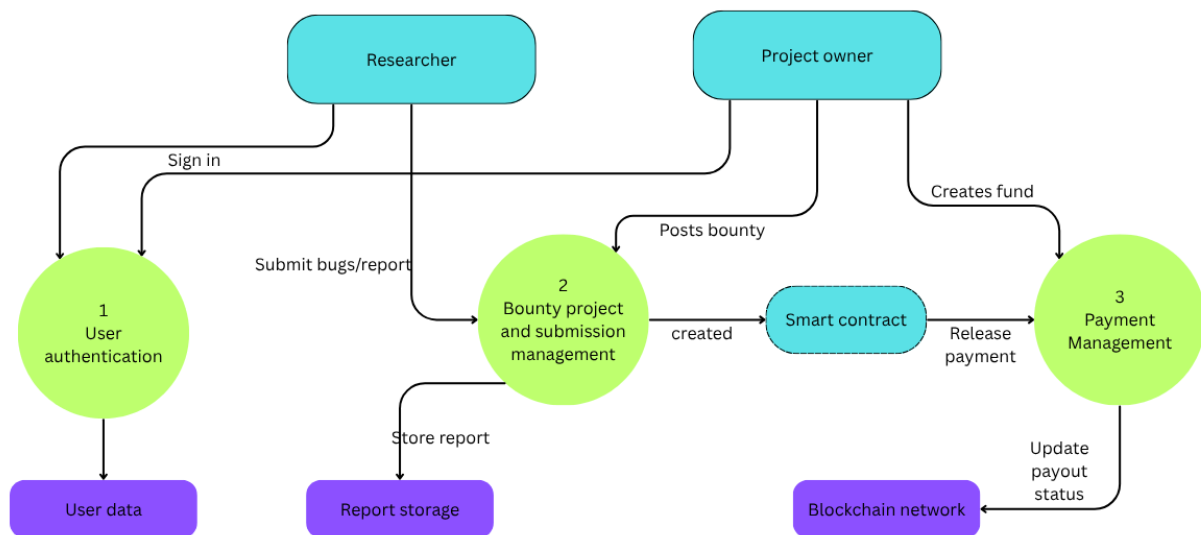


Fig No. 04 - DFD Level-1 Diagram

Flowchart:

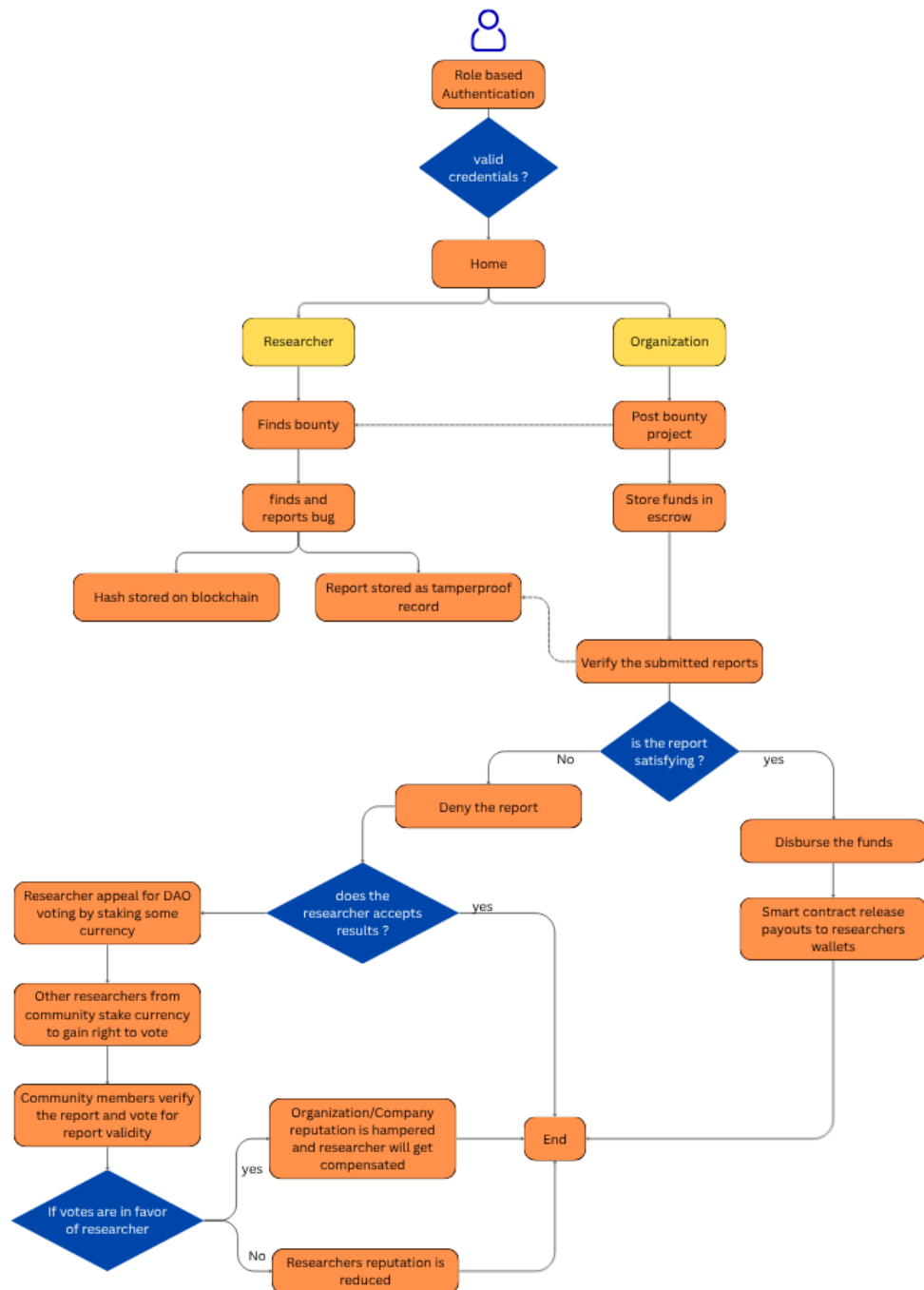


Fig No. 05 - Flowchart

4.4 Project Scheduling & Tracking using Time line / Gantt Chart

The DeBug Project Timeline 2025 – Gantt Chart represents the structured schedule of development phases, major tasks, and their durations from August to November 2025. The project begins with the Planning Phase, which includes topic selection, literature review, requirement gathering, and synopsis submission. Following this, the Initial Development Phase focuses on setting up the frontend interface and backend framework using React.js, Node.js, and Express.js. After the first review, the Core Development Phase involves implementing Solana smart contracts, IPFS integration, and MetaMask-based authentication. The subsequent Advanced Development Phase concentrates on establishing report encryption, smart contract-triggered payouts, and the backend validation workflow using Express.js and Prisma ORM. In the later stages, the Testing and Debugging Phase ensures proper synchronization between on-chain and off-chain components, validating smart contract performance and data flow. The Final Review Phase emphasizes user interface refinement, analytics dashboard integration, and final report preparation. The timeline indicators confirm that most modules — including user management, submission handling, payment automation, and governance — have been successfully implemented or are in their final testing stages, showcasing a well-structured, timely, and systematic execution of the project.

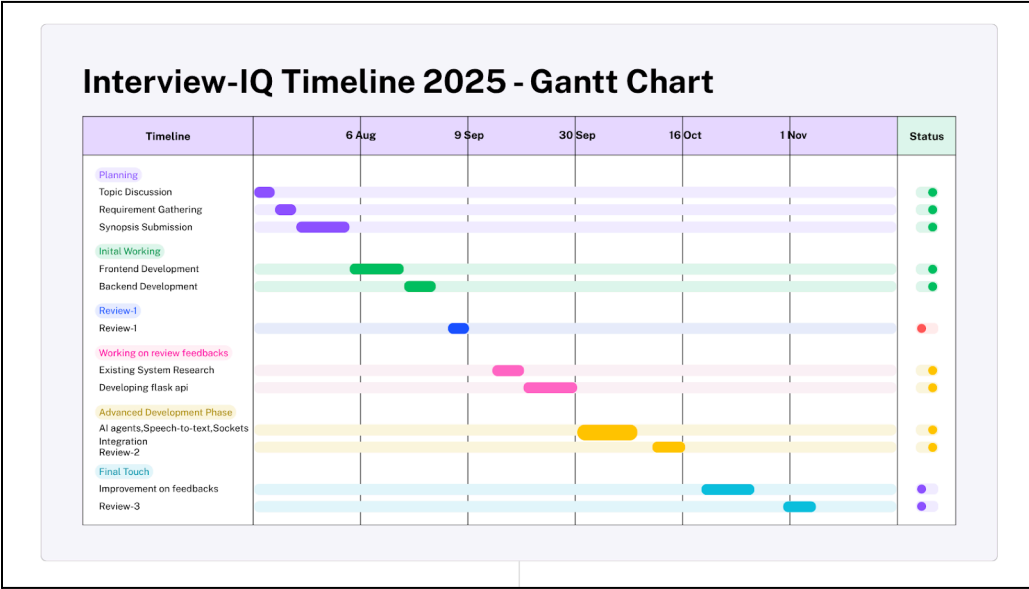


Fig No. 06 - Gantt Chart

5. Results and Discussions

The development of DeBug has successfully achieved major milestones, with most of the core modules implemented, tested, and functionally integrated. The frontend, built using React.js and Tailwind CSS, provides a clean and intuitive interface that enables smooth navigation across modules such as bounty creation, submission tracking, profile management, and analytics dashboards. The design emphasizes usability and transparency, ensuring that both researchers and organizations can interact seamlessly with the system.

On the backend, services have been developed using Node.js and Express.js, facilitating secure data exchange between the client and blockchain network. Key functionalities such as MetaMask-based authentication, bounty configuration, and encrypted report submission have been fully implemented and tested for stability. The integration of PostgreSQL ensures reliable storage of user, bounty, and transaction metadata, while IPFS is used for decentralized report storage to maintain immutability and privacy.

A dedicated Solana smart contract has been successfully deployed to manage escrow creation, fund locking, and automated reward disbursement. Upon validation of a vulnerability, the contract triggers instant payouts to the researcher's wallet, confirming the system's trustless and transparent financial flow. The Express-Prisma backend efficiently handles validation and triage, linking on-chain transactions with off-chain operations to preserve performance without compromising security.

In terms of performance, smart contract execution and API response times have been optimized to minimize latency, ensuring smooth coordination between on-chain and off-chain layers. The reputation module has been implemented successfully, allowing researchers to maintain verifiable credibility across multiple programs. Meanwhile, the DAO governance framework for community-driven triage and dispute resolution has been partially integrated, marking significant progress toward decentralized decision-making.

The analytics module provides clear visual dashboards for tracking bounties, submissions, and payments, with real-time updates reflecting transparency across all processes. Furthermore, security and privacy tests have confirmed the integrity of encrypted submissions and user identity masking through Zero-Knowledge (ZK) verification mechanisms.

Overall, the DeBug system demonstrates strong proof of concept, successfully combining blockchain transparency with practical off-chain functionality. With all primary modules—authentication, submission handling, smart contract payouts, and analytics—operational, the platform offers a secure, decentralized, and scalable environment for ethical vulnerability disclosure. Future work will focus on enhancing DAO governance participation, integrating automated severity scoring, and refining user analytics visualization to further strengthen system usability and researcher engagement.

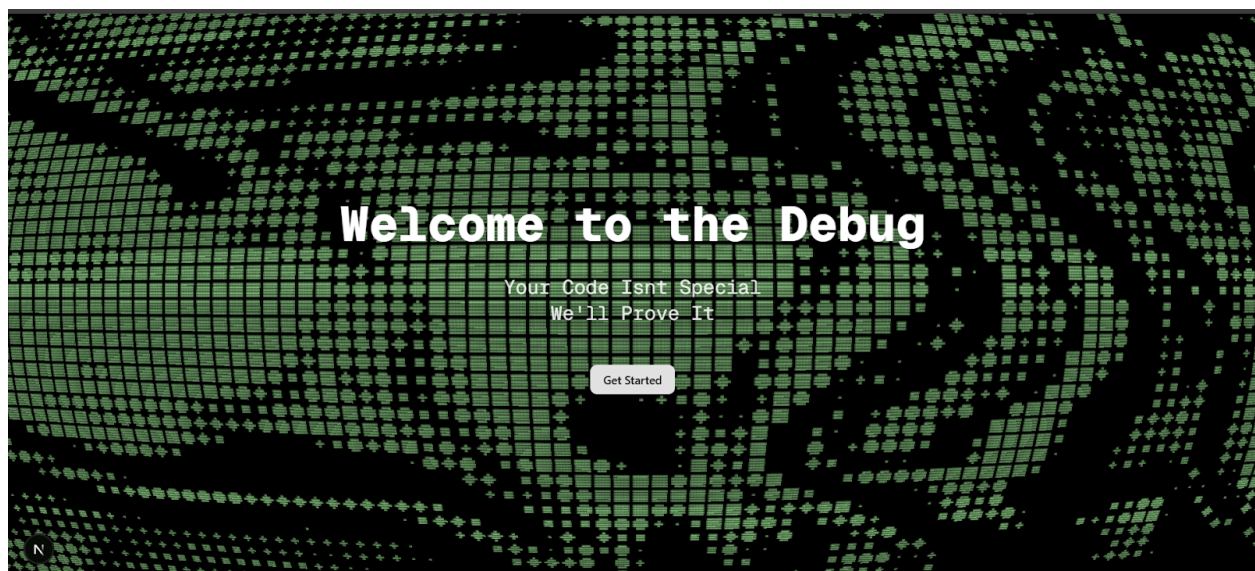


Fig No. 07 - Get Started Page

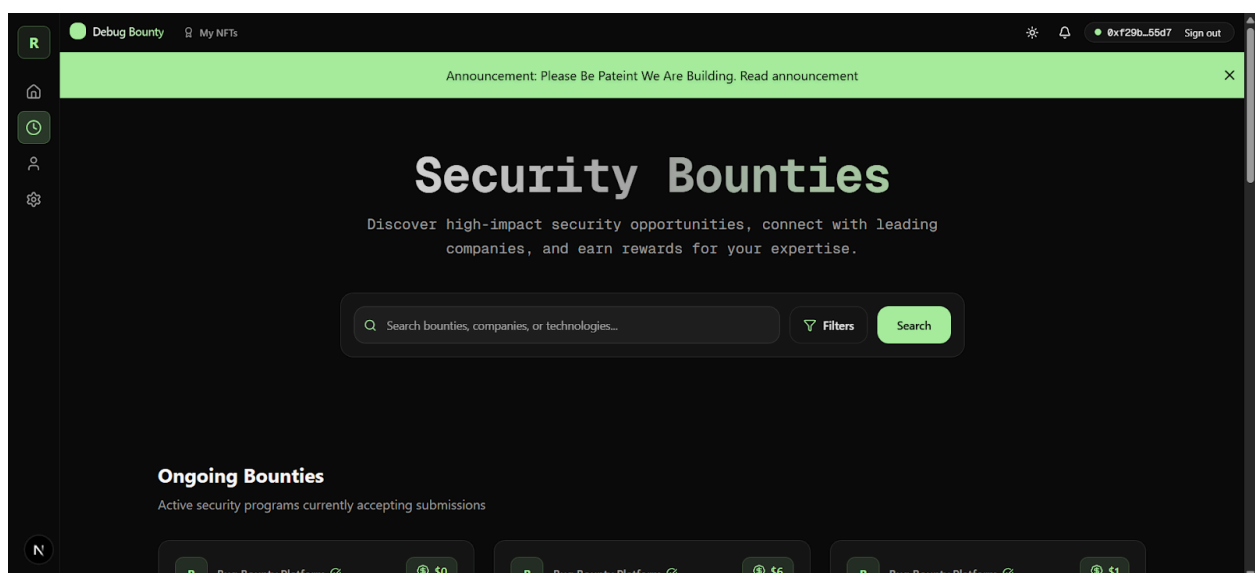


Fig No. 08 - Home Page

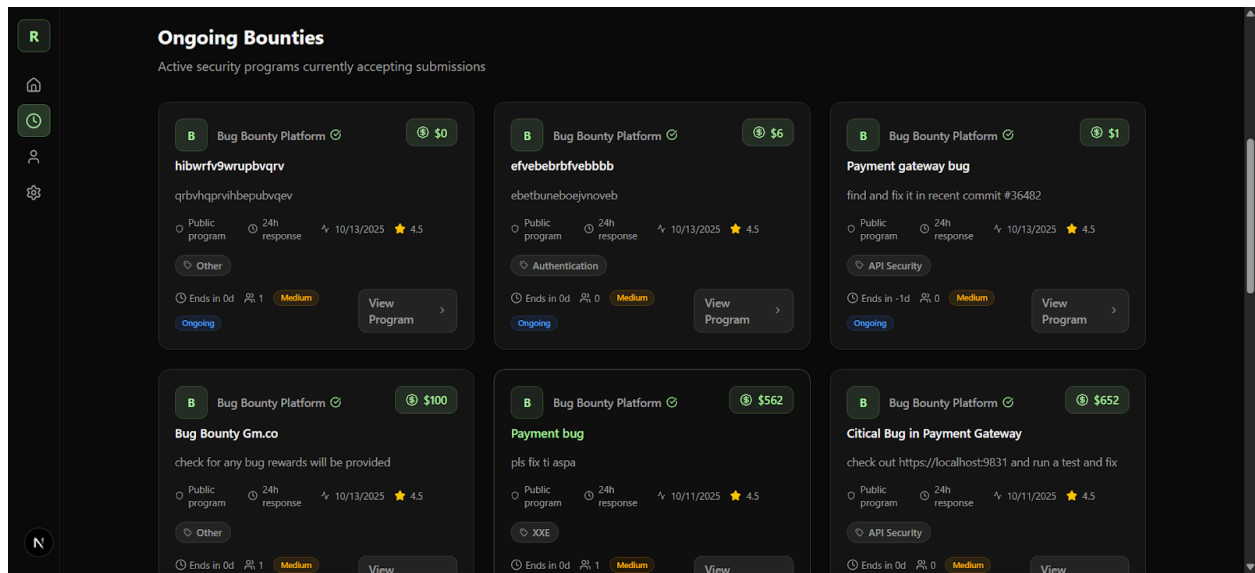


Fig No. 09 - Ongoing Bounty Checking Page

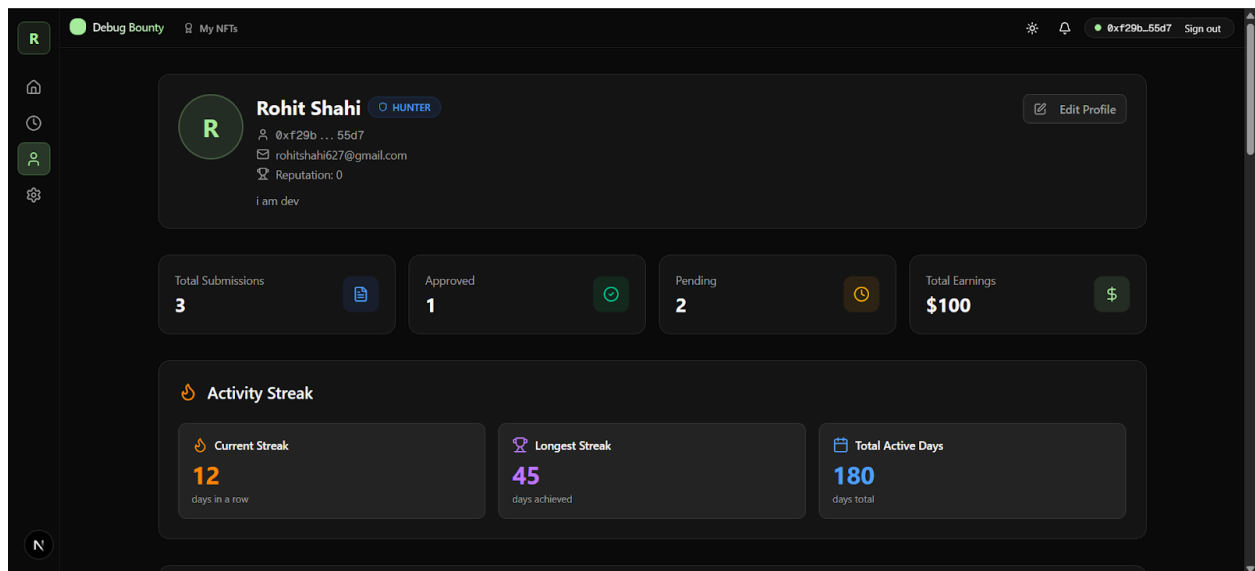


Fig No. 10 - View Profile Page

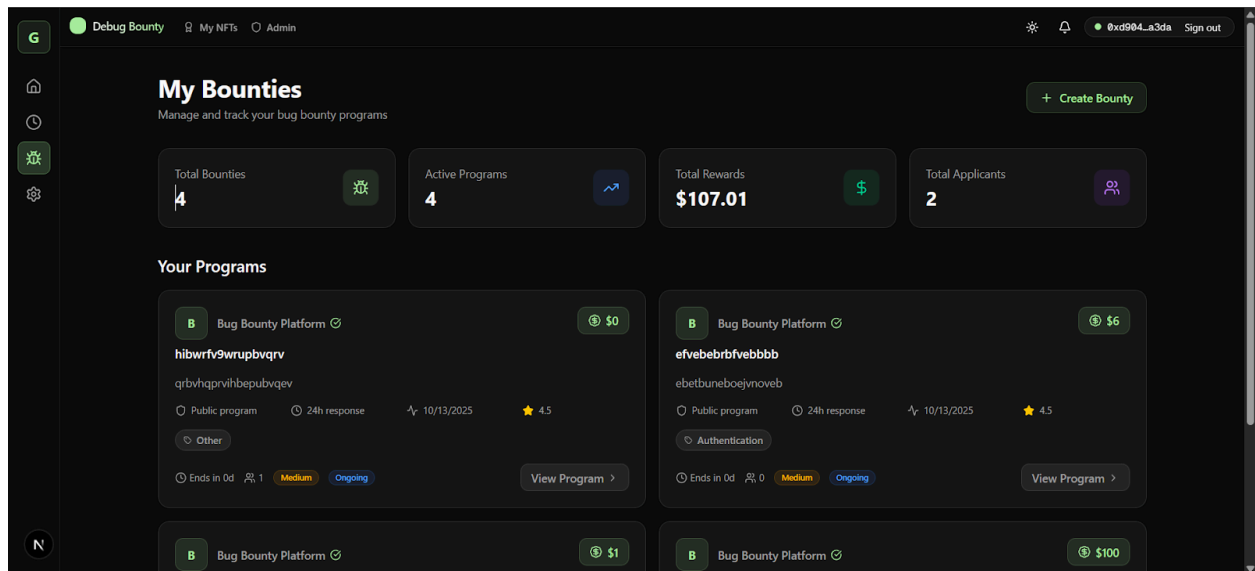


Fig No. 11 - My Bounties Page

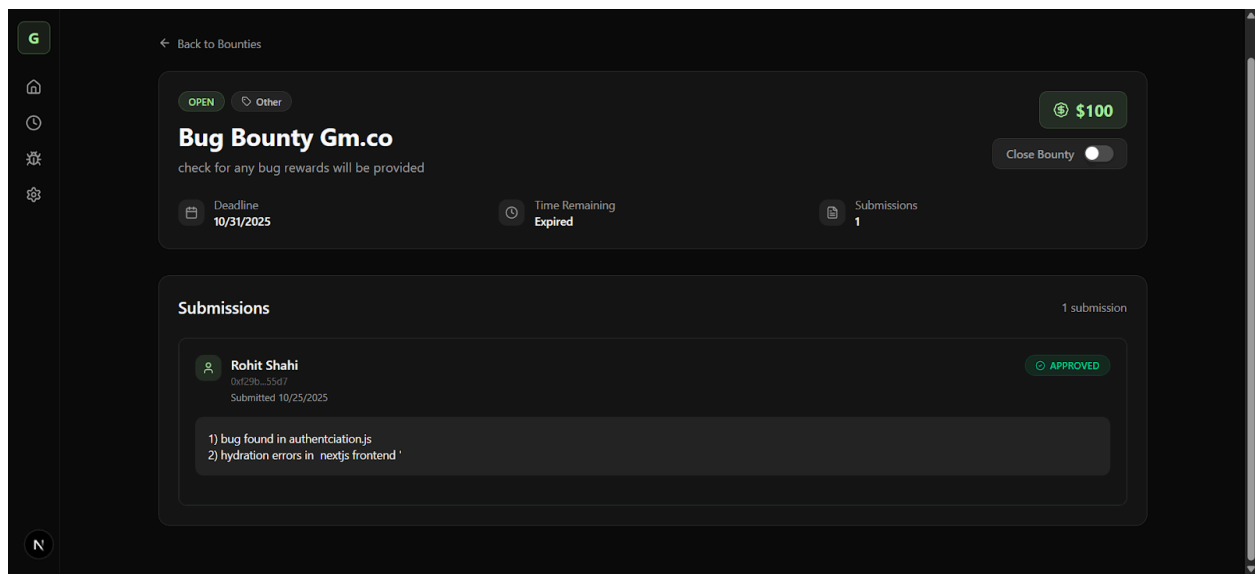


Fig No. 12 - Submissions and Deadline Page

6. Plan of action for the next semester

In the upcoming development phase, the main focus for DeBug will be on strengthening decentralization, automation, and scalability to prepare the system for real-world deployment. The first priority will be implementing the DAO governance module, allowing community members to participate in vulnerability triage and dispute resolution through decentralized voting. This addition will ensure transparent decision-making and eliminate bias from centralized moderation.

The reputation system will be enhanced with a dynamic scoring mechanism that evaluates researchers based on submission accuracy, response time, and consistent participation. This upgrade aims to promote credibility, reward ethical behavior, and improve collaboration between researchers and organizations.

On the blockchain side, smart contracts on Solana will be optimized to support multiple token payouts, gas-efficient transactions, and automated reward distribution based on vulnerability severity. These improvements will make the reward process faster, fairer, and more cost-effective.

The analytics and monitoring module will be upgraded to include real-time dashboards with visual reports on submissions, payments, and researcher performance. Integrated WebSocket-based notifications will provide instant updates regarding report status, triage outcomes, and payouts, ensuring transparent communication.

For enhanced privacy, Zero-Knowledge (ZK) identity verification will be extended to support selective disclosure, allowing researchers to prove authenticity without revealing personal data. Additionally, the backend performance will be optimized using Redis caching for faster data access and reduced API latency.

Lastly, all backend and blockchain services will be Dockerized and deployed on AWS ECS / Kubernetes to ensure high availability, scalability, and simplified maintenance. Improved IPFS integration will enable decentralized, encrypted storage of vulnerability reports and related evidence. Together, these enhancements will make DeBug a transparent, efficient, and community-driven decentralized platform for secure vulnerability management.

7. Conclusion

The development of DeBug – A Decentralized Bug-Bounty Platform marks a significant step toward transforming the cybersecurity ecosystem through blockchain-based transparency, automation, and privacy preservation. By combining smart contract-powered escrows, IPFS-based decentralized storage, and wallet-based authentication, the system successfully demonstrates a trustless and verifiable framework for ethical vulnerability disclosure.

The current stage of development has established a strong foundation, with fully functional modules for user authentication, bounty creation, encrypted report submission, and automated payout execution. The hybrid on-chain/off-chain model ensures scalability while maintaining the integrity and immutability of on-chain transactions.

With the next phase focused on DAO governance integration, advanced reputation tracking, improved analytics dashboards, and automated severity-based rewards, DeBug is well-positioned to evolve into a comprehensive decentralized solution for ethical hacking and vulnerability management. In the long term, the project aims to redefine digital trust by enabling secure, transparent, and equitable collaboration between researchers and organizations, ultimately strengthening global cybersecurity resilience.

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9. Appendix

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